

Manufacturing Cybersecurity Risk Assessment

Identify cyber risks in a connected manufacturing cell and recommend layered safeguards.

Customer Aegis Aerospace	Department Operational Technology Security	Difficulty Intermediate	Time 1-2 class periods
------------------------------------	---	-----------------------------------	----------------------------------

Production Goal: Reduce cyber risk in a connected manufacturing workcell.

Background

A connected aerospace factory experienced unusual robot commands after a maintenance laptop was attached to the network. Production was stopped before damage occurred. Your team must assess the workcell using confidentiality, integrity, and availability principles.

Learning Objectives

- Apply the CIA triad to manufacturing systems
- Identify technical and human vulnerabilities
- Prioritize risk using likelihood and impact
- Recommend layered controls

Equipment

- Workcell/network diagram
- Risk matrix template
- Scenario cards
- Engineering notebook

Student Instructions

- 1 Review the provided workcell diagram and identify all devices, connections, and users.
- 2 List at least ten assets or data types that require protection.
- 3 Identify at least eight vulnerabilities, including one human-factor risk.
- 4 For each risk, score likelihood and impact from 1 to 5.
- 5 Plot risks on a matrix and select the four highest priorities.
- 6 Recommend preventive, detective, and recovery controls for each priority.
- 7 Create an incident response sequence for an unexpected robot command.
- 8 Present the top risks and explain how each relates to confidentiality, integrity, or availability.

Engineering Requirements

- Identify at least eight distinct vulnerabilities
- Include risks to safety and production availability
- Recommend at least two controls for each top risk
- Include an isolation and recovery procedure

Constraints & Safety

- Do not test real malware or bypass security controls

- Use only provided diagrams and scenarios
- Recommendations must preserve authorized maintenance access
- Distinguish safety stop from cybersecurity response

Deliverables

- Annotated system diagram
- Risk register and matrix
- Top-four mitigation plan
- Incident response flowchart
- Reflection

Reflection Questions

- 1 Which risk had the highest safety impact?
- 2 How can data integrity affect physical quality?
- 3 Why are backups not sufficient prevention?
- 4 What training would reduce human risk?

Engineering Support

Engineering Tip

Use layered defense: one control may fail, so combine access control, monitoring, segmentation, and recovery.

Common Mistake

Cybersecurity is not only confidentiality; altered coordinates can create a physical hazard.

Stretch Goal

Design a network-segmentation diagram separating office, maintenance, and production systems.

Career Connection

Operational technology cybersecurity specialists protect robots, PLCs, sensors, and industrial networks.

Before You Submit

- Notebook evidence included
- Requirements checked
- Testing documented
- Reflection completed